

Performance Test Summary — Project Pegasus Sprint 3

Field	Value
Date	2026-04-09
Sprint	Sprint 3 (Mar 31 — Apr 9)
Prepared by	QA Team
Status	Partial — k6 tests not yet executed

1. Test Tooling

Tool	Purpose	Status
Playwright	API timing — measures response latency during end-to-end test execution	Executed
curl	Endpoint latency — manual spot checks on individual endpoints	Executed
k6	Load testing suite — synthetic load generation with configurable virtual users and scenarios	Written, NOT executed this sprint

2. k6 Test Suite

Location

UBN/load-tests/k6/

Configured Scenarios

Scenario	Target	Virtual Users	Duration
Aggregate throughput	500 TPS across all endpoints	100	5 minutes
P99 latency	< 200ms per request	50	3 minutes
Sustained load	Steady-state 200 TPS	50	15 minutes
Spike test	0 to 1000 TPS ramp	200	2 minutes ramp, 3 minutes hold

Why Not Executed

The k6 test suite was written during Sprint 3 but could not be executed because 9 of the 16 services in the dev environment are currently unreachable. Load testing against an incomplete environment would produce invalid results. Execution is deferred until the dev environment is fully operational.

3. Measured Latencies (Playwright)

The following latencies were captured from Playwright test execution against the dev environment. Each measurement represents the wall-clock time from request initiation to response receipt.

3.1 Health Endpoints

Service	Endpoint	Latency	Verdict
---------	----------	---------	---------

Admin Gateway	GET /health	~200ms	PASS
Admin Auth	GET /health	~200ms	PASS
User Management	GET /health	~200ms	PASS

Health endpoints are lightweight and respond within acceptable limits. The ~200ms latency includes network round-trip time to the VPS.

3.2 Sandbox Mock Endpoints

Service	Endpoint	Latency	Verdict
Partner Gateway	POST /api/v1/payments/account-enquiry (sandbox)	27-30 seconds	FAIL
Partner Gateway	POST /api/v1/kyc/bvn/verify (sandbox)	27-30 seconds	FAIL
Partner Gateway	POST /api/v1/accounts/balance (sandbox)	27-30 seconds	FAIL

ISSUE: PEG-381 — Severe Performance Degradation on Sandbox Mock Endpoints

All sandbox mock endpoints are exhibiting response times of 27 to 30 seconds, far exceeding the 200ms target. This is a critical performance issue that renders the sandbox environment unusable for partner integration testing.

Suspected root causes:

- Database connection pool exhaustion on the Bank Adapter service.
- Synchronous blocking on the mock HTTP client used for sandbox responses.
- DNS resolution delays on the VPS for outbound mock service calls.

3.3 Partner Portal

Page	Metric	Value	Threshold	Verdict
Login page	Full page load	2.2 seconds	< 5 seconds	PASS
Dashboard	Full page load	2.6 seconds	< 5 seconds	PASS
API Keys page	Full page load	2.4 seconds	< 5 seconds	PASS

The Partner Portal is a Next.js application served behind Cloudflare CDN. Page load times are within the 5-second acceptance threshold.

3.4 Admin Portal

Page	Metric	Value	Threshold	Verdict
Login page	DOM interactive	~966ms	< 3 seconds	PASS
Login page	Server response (TTFB)	375ms	< 1 second	PASS
Dashboard	Full page load	~1.2 seconds	< 5 seconds	PASS

The Admin Portal demonstrates good performance with sub-second server response times.

3.5 Admin API Endpoints

Endpoint	Method	Latency	Threshold	Verdict
/api/admin/partners	GET	~800ms	< 2 seconds	PASS

/api/admin/partners/{id}	GET	~450ms	< 2 seconds	PASS
/api/admin/production-access-requests	GET	~1.1 seconds	< 2 seconds	PASS
/api/admin/production-access-requests/{id}/approve	POST	~1.5 seconds	< 2 seconds	PASS

All Admin API endpoints respond within the 2-second threshold.

3.6 User Management Validation

Endpoint	Method	Latency	Threshold	Verdict
/api/Auth/register (validation error)	POST	~240ms	< 1 second	PASS
/api/Auth/verifyEmailCode (invalid code)	POST	~220ms	< 1 second	PASS

Validation responses are fast, indicating the request pipeline short-circuits correctly on invalid input without hitting the database.

4. Performance Issues Found

PEG-381: Sandbox Mock Endpoints — 27-30 Second Latency

Property	Value
Jira ID	PEG-381
Severity	Critical
Affected endpoints	All sandbox mock endpoints via Partner Gateway
Expected latency	< 200ms
Actual latency	27,000 — 30,000ms
Degradation factor	~135x slower than target
Status	Open

Impact: Partners cannot test integrations against the sandbox environment. The 30-second response time will cause client-side timeouts in most HTTP clients (default timeout is typically 30 seconds).

Recommended investigation steps:

- 1. Check Bank Adapter service container logs for timeout or connection errors.
- 2. Inspect database connection pool metrics on the Bank Adapter.
- 3. Verify DNS resolution time from the VPS to external mock endpoints.
- 4. Check for synchronous blocking calls in the sandbox mock handler.

5. Non-Functional Requirement (NFR) Targets

NFR	Target	Status
Throughput	500 TPS aggregate	NOT YET VALIDATED
Latency (P99)	< 200ms	NOT YET VALIDATED
Uptime	99.9%	NOT YET VALIDATED
Page load (Partner Portal)	< 5 seconds	PASS (measured 2.2-2.6s)

Page load (Admin Portal)	< 5 seconds	PASS (measured ~1.2s)
Health endpoint response	< 1 second	PASS (measured ~200ms)

The primary NFR targets (500 TPS, P99 < 200ms, 99.9% uptime) have **not yet been validated** because the k6 load tests were not executed during this sprint. These require a fully operational dev environment with all 16 services running.

6. Recommendation

Execute k6 load tests once the dev environment is fully operational.

Specific prerequisites:

1. All 9 unreachable banking services must be brought online.
2. The Partner Gateway 502 issue must be resolved.
3. PEG-381 (sandbox mock latency) must be investigated and fixed.
4. Database connection pools should be configured with appropriate limits before load testing.

Once these prerequisites are met, the k6 test suite in `UBN/load-tests/k6/` can be executed against the dev environment to validate the 500 TPS, P99 < 200ms, and 99.9% uptime targets.

7. Cloudflare CDN Caching Observations

During testing, the following CDN caching behavior was observed on frontend services:

Header	Value	Meaning
cf-cache-status	DYNAMIC	Cloudflare is not caching the response (dynamic content)
x-nextjs-cache	HIT	Next.js internal cache served the page from its cache layer

Interpretation: Cloudflare is treating all responses as dynamic (no edge caching). However, Next.js is caching pages at the application layer, which explains the good page load times despite the lack of CDN caching. For production, enabling Cloudflare edge caching for static assets and ISR pages would further improve performance.

8. Summary

Area	Result
Health endpoints	PASS (~200ms)
Sandbox mock endpoints	FAIL (27-30s — PEG-381)
Partner Portal pages	PASS (2.2-2.6s)
Admin Portal pages	PASS (~966ms DOM, 375ms TTFB)
Admin API endpoints	PASS (all < 2s)
User Management validation	PASS (~240ms)
k6 load tests (500 TPS, P99)	NOT EXECUTED
NFR validation	INCOMPLETE

Sprint 3 performance testing is partial. The endpoints that are reachable perform well (with the critical exception of sandbox mocks). Full NFR validation requires the k6 suite to be executed in Sprint 4 once the environment is stable.